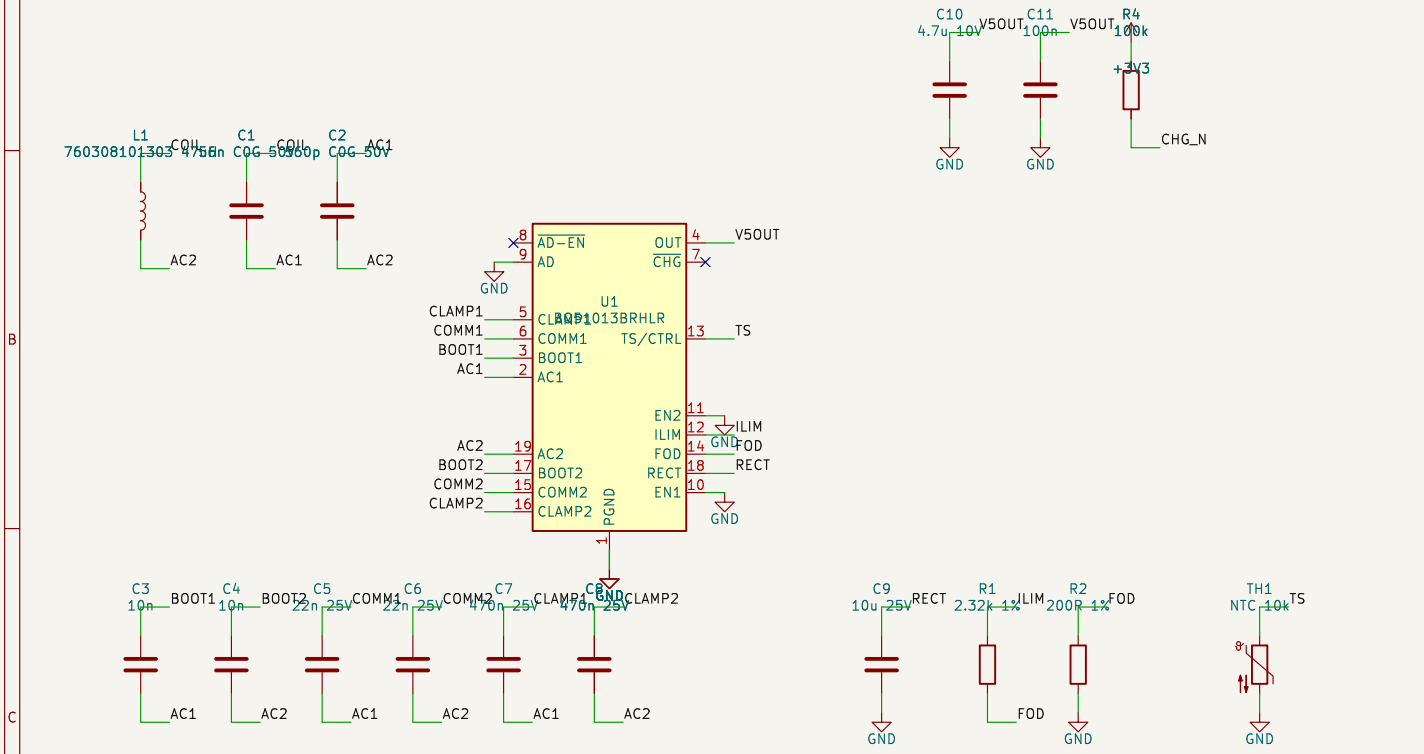
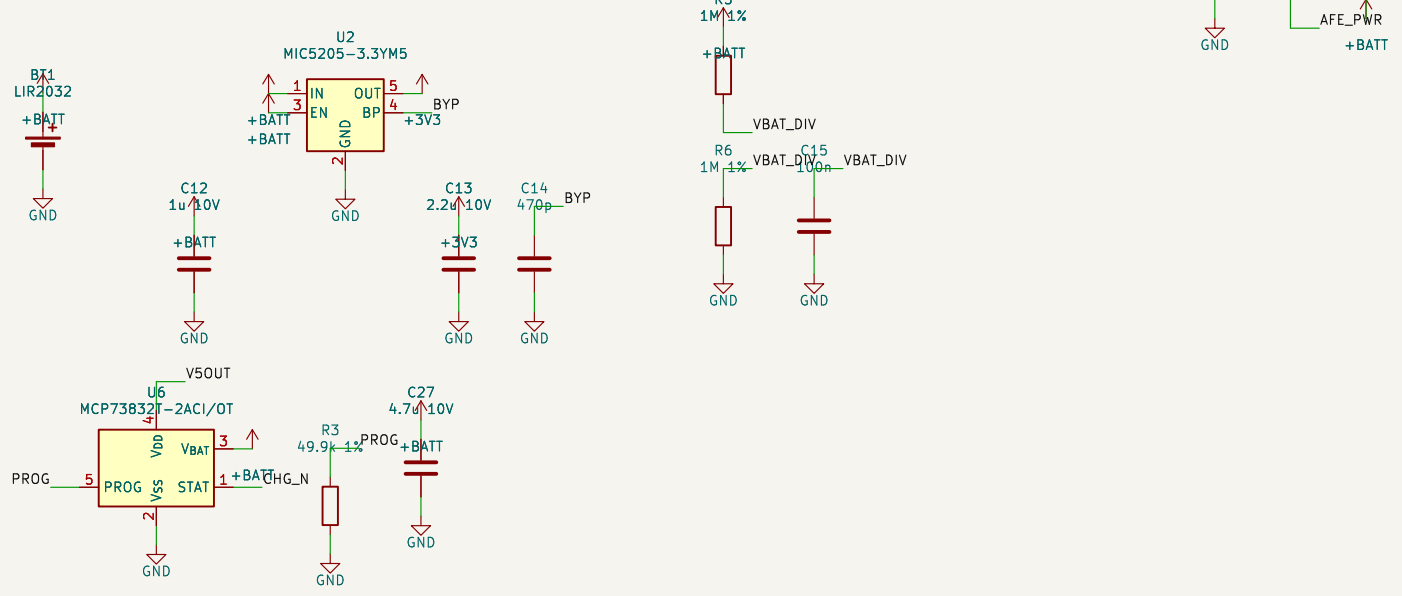


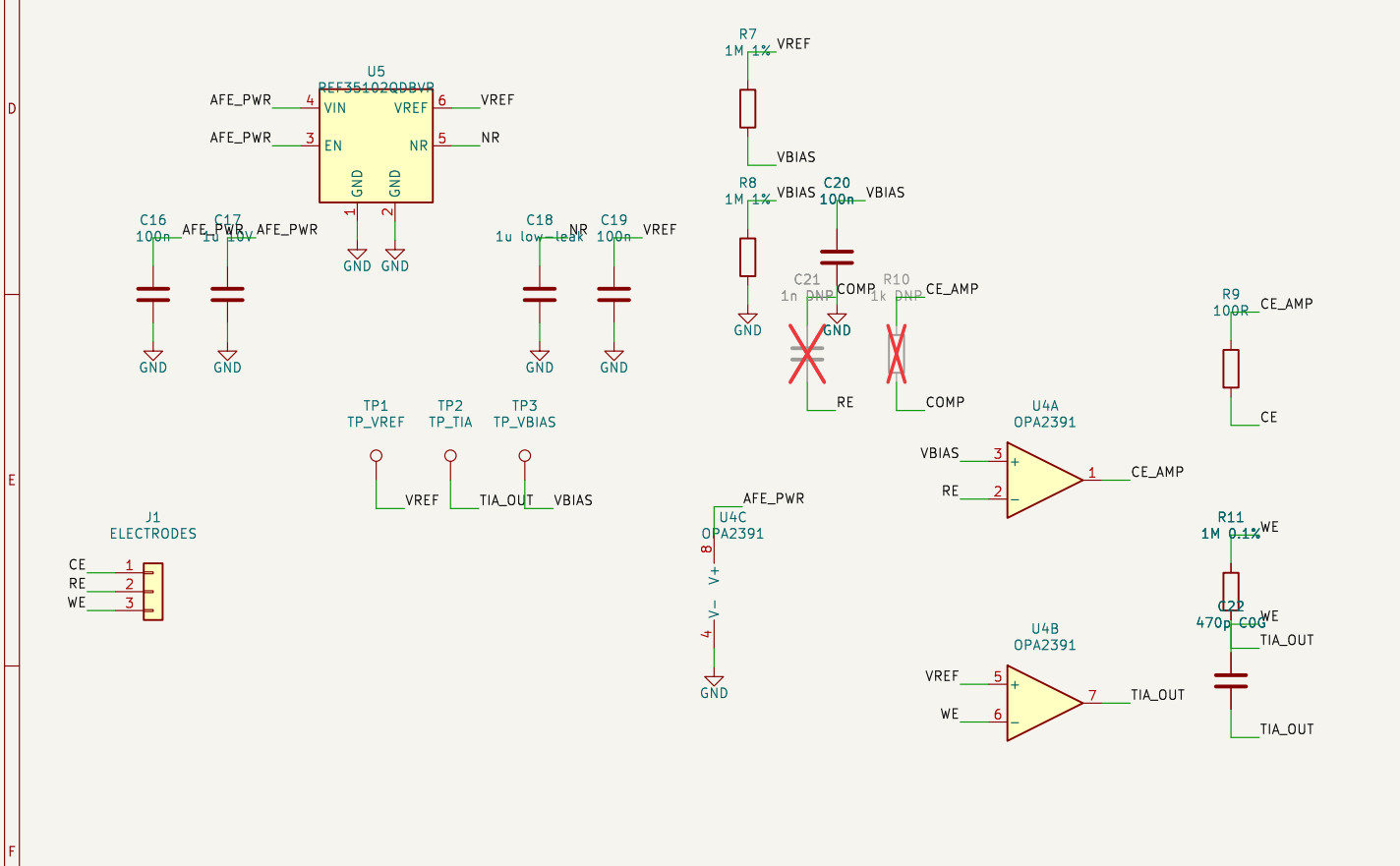
QI RECEIVER 5V (BQ51013B) + 20mA CHARGER (MCP73832)



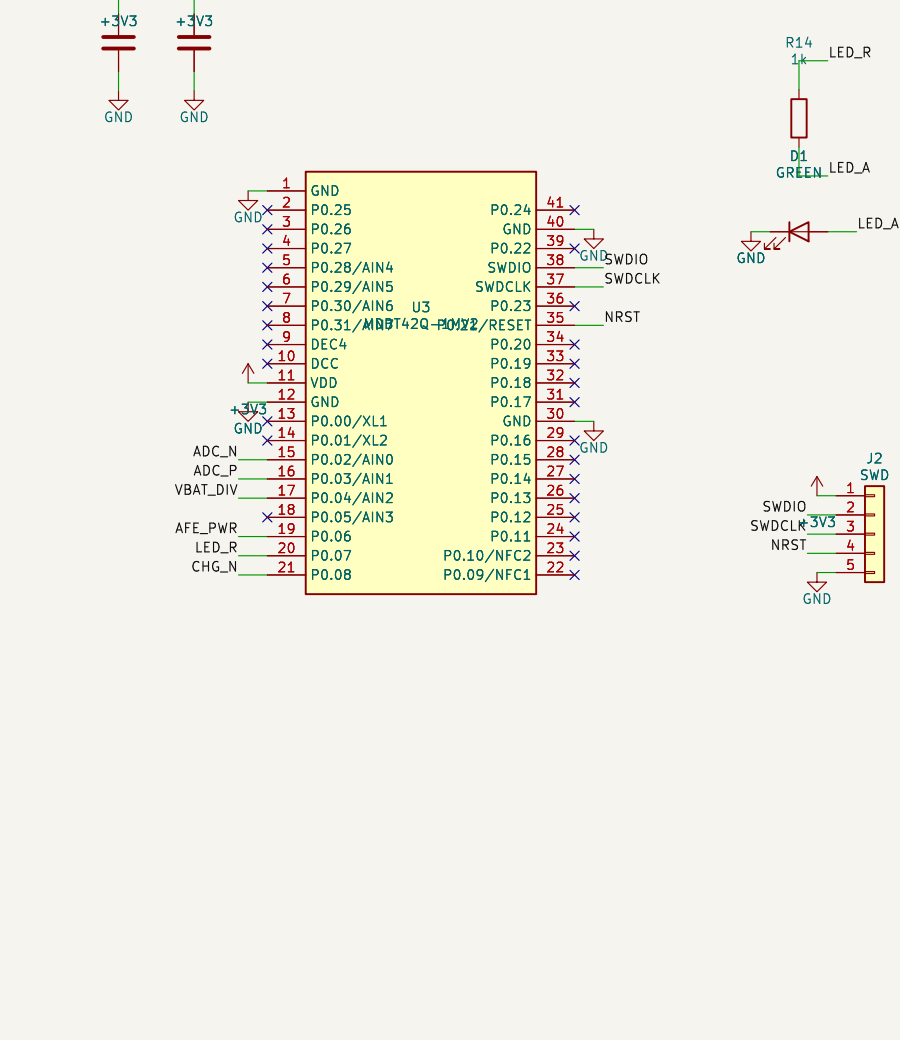
BATTERY + 3.3V LDO + VBAT MONITOR



ANALOG FRONT-END: 2-OP-AMP POTENTIOSTAT (+0.512V)



nRF52832 MODULE (MDBT42Q) + SWD + LED



DISCRETE POTENTIOSTAT – wireless–charged 3–electrode chronoamperometry
 $E_{cell} = WE - RE = 1.024V - 0.512V = +0.512V$ fixed. $I_{WE} = (V(ADC_P) - V(ADC_N)) / RF$

- DESIGN NOTES / VERIFY BEFORE LAYOUT
- CHARGING: two–stage per TI guidance for small cells. BQ51013B = Qi receiver with regulated 5V (V5OUT). ILIM: RILIM = 250/IMAX = R1+R2 = 2.52k $\rightarrow$ IMAX 100mA (HW limit 120mA). MCP73832–2 (4.2V) charges the LIR2032: ICHG = 1000/RPROG = 1000/49.9k = 20mA (0.5C). termination $\sim 7.5\% = 1.5mA$, open–drain STAT $\rightarrow$ CHG_N (100k pullup to 3V3). This replaces the earlier BQ51050B direct–charge (TI: not recommended <200mA).
 - RESONANT TANK: L1 47uH (Würth 760308101303, dia 26.3mm). $C1 = 1 / (L * (2\pi * 100kHz)^2) = 54n \rightarrow 56n$ COG. C2 $\rightarrow$ fd = 1MHz $\rightarrow$ 560p COG. Retune both whenever the coil changes.
 - MDBT42Q pads 22–36 are a sequential reconstruction (P0.09/NFC1 ... P0.23); pads 1–21 and 37–41 verified against the datasheet. Every pad actually used in this design is a verified one. Cross–check the full pad table against the Raytac datasheet before PCB layout.
 - TS/CTRL: 10k NTC against the LIR2032 holder cuts wireless power if the cell heats (plain 10k = sense defeated, still a valid safe strap). EN1=EN2=GND: wireless enabled.
 - AFE_PWR is driven by nRF GPIO P0.06 set to high–drive. Load = REF35+OPA2391 ~ 60uA. Power the AFE off between runs to avoid electrode polarization.
 - WE net: guard ring driven from VREF, no soldermask opening at TIA input, clean flux.
 - RF/CF options: 1M/470p (default, +2.1uA/~0.9uA FS), 5.1M/100p (427/181nA), 10M/47p (217/92nA).
 - R10h/C21 control–amp compensation are DNP: fit only if the CE–RE loop oscillates.
 - LIR2032 has no protection circuit: firmware must System–OFF below 3.0V (VBAT_DIV, AIN2).
 - SAADC: differential P=AIN1(ADC_P) N=AIN0(ADC_N), gain 1/2..4, oversample 64x default.